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Multi-Head Networks for Missing Modalities: A Study on Robustness in Multimodal Learning

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Abstract. Multimodal learning has emerged as a critical area of research in artificial intelligence, enabling models to integrate and process diverse data sources such as images, text, audio, and sensor readings. However, real-world multimodal systems are often challenged by missing modalities due to sensor failures, occlusions, or data transmission issues, leading to degraded performance. To address this, we investigate the robustness of Multi-Head Networks (MHNs) trained with an orthogonality component in the loss function, which enforces decorrelation among latent representations to enhance feature diversity and resilience to missing data. We develop and evaluate three MHN architecture variants, each leveraging different feature extraction strategies. Our experiments, conducted on a recently published multimodal dataset for behavioral biometrics, include an ablation study in which we systematically remove the most informative features and assess the impact of the orthogonality constraint. Results show that CNN-based architectures benefit the most from orthogonality loss, FNNs show moderate gains, while LSTM networks exhibit a variable response, suggesting that the effectiveness of this constraint is model-dependent. These findings also highlight the need for architecture-specific robustness strategies.

Keywords. Multi-Head Deep Learning, Multimodal AI, Robust AI, High Quality AI

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1. Introduction and literature review

Multimodal learning is an essential area of Artificial Intelligence (AI), enabling models to integrate and process diverse data sources such as images, text, audio, and sensor readings. These models are widely applied in fields including autonomous driving [1], healthcare [2], emotion recognition [3], and human-computer interaction [4,5]. Multimodality mirrors the human ability to synthesize information from multiple sensory inputs, enhancing perception and interaction.

Despite its advantages, multimodal learning faces significant challenges in real-world applications due to missing data or missing modalities, which can arise from sensor failures, occlusions, transmission errors, or adversarial conditions. The absence of critical modalities during training or inference can severely degrade model performance. To mitigate this, recent advances in Multimodal Learning with Missing Modalities (MLMM) have explored techniques to enhance model robustness [6].

Various approaches have been proposed to address these challenges, including shared-specific feature modeling to separate shared and modality-specific information [7], parameter-efficient adaptation to refine pre-trained models with minimal overhead [8], and Bayesian meta-learning to enable adaptation in extreme missing modality scenarios [9]. Another emerging method is multimodal prompt learning, which uses modality-aware prompts to guide inference and maintain accuracy even with missing inputs [10][11].

These studies highlight the importance of assessing the robustness of multimodal models in the presence of incomplete inputs. In this context, robustness is a defining trait of high-quality AI, reflected not only in performance under ideal conditions, but in the model's ability to remain reliable when faced with data variability or degradation.

One promising direction to support such robustness lies in architectural designs that can capture complementary information across modalities, while also exploiting redundancy to compensate for potential data loss. In this context, Multi-Head Networks (MHNs) provide a flexible framework in which multiple parallel processing heads independently extract diverse feature representations. Originally introduced in the context of *self-attention* mechanisms [12], they have since become a core component in various multimodal applications, including vision-language modeling [13], audio-visual event localization [14], and cross-modal retrieval [15]. Beyond attention-based models, multi-head architectures have also been successfully applied in multi-task learning [16], object detection [17], and other deep learning frameworks to promote modularity, feature diversity, and generalization.

In this study, we extend the MHN paradigm by designing architectures in which each head is an independently parameterized sub-network operating on the same input. To promote representational diversity and robustness, we introduce a novel training loss that incorporates an orthogonality-based regularization term. This term encourages decorrelation among the latent representations produced by different heads, with the aim of enhancing robustness to missing information.

We tested three architecture variants, *m-FFNet*, *m-ConvNet*, and *m-LSTMNet*, each corresponding to a different feature extraction strategy. In all cases, the prefix *m* denotes the number of parallel and independently trained heads, which function as sub-networks within the larger MHN architecture. A comparative ablation study is conducted to assess the impact of the orthogonality loss against standard loss functions when key modality features are missing.

Our results indicate that the proposed framework is both conceptually sound and practically effective. Models trained with orthogonality constraints consistently demonstrate improved robustness under feature ablation, with the convolutional configuration (*m-ConvNet*) yielding the most pronounced gains.

The remainder of this paper is structured as follows. Section 2 details the proposed methodology, including model architectures and the loss function design. Section 3 presents the experimental setup, ablation study, and obtained results. Finally, Section 4 discusses conclusions, limitations, and future directions.

2. Methodology

This section describes the methodology employed in designing and evaluating the proposed models and loss function. The work integrates multiple neural network architectures to extract features from input data and applies a novel loss function to improve classification performance through an orthogonality constraint on latent representations. The new orthogonality loss is specifically designed to mitigate performance drops in scenarios where certain inputs are missing. While the primary task remains classification, the addition of the orthogonality loss enhances the robustness of the model against input perturbations, ensuring more stable feature representations.

The idea of enforcing orthogonality to promote feature diversity or separation has been explored in previous works for other purposes [18,19]. However, unlike these prior studies, our orthogonality constraint is specifically adapted to multi-head architectures to promote resilience against missing modalities in multimodal learning.

2.1. Model Architectures

We design a neural network architecture composed of m base blocks that perform feature extraction and output latent representations of the input space. The m latent vectors are concatenated and passed to a fully connected layer, which uses them to predict the logits. The logits are utilized in the classification loss, while the latent vectors are employed in the orthogonality loss.

The architectures explored in this study differ based on the choice of feature extraction base blocks. Specifically, we design the following models:

- *m-FFNet*: Combines *m FFNet* blocks.
- *m-ConvNet*: Combines *m ConvNet* blocks.
- *m-LSTMNet*: Combines *m LSTMNet* blocks.

The feature extraction base blocks are described as follows:

- Feedforward Networks (*FFNet*): A simple feedforward network consisting of two fully connected layers [20] with a ReLU activation function applied after the first layer. This base block is used in the m-FFNet architecture.
- Convolutional Neural Networks (*ConvNet*): A CNN-based [21] architecture consisting of two convolutional layers, adaptive average pooling, and a fully connected layer. This base block is used in the m-ConvNet architecture.
- LSTM Networks (*LSTMNet*): A recurrent neural network block composed of an LSTM [22] layer followed by a fully connected layer. This base block is used in the m-LSTMNet architecture.

2.2. Loss Function Design

The total loss function defined in Eq. 1 is used to train the model and is a weighted combination of the classification loss and the orthogonality loss:

$$\text{Total Loss} = \alpha \cdot \text{Classification Loss} + \beta \cdot \text{Orthogonality Loss}, \tag{1}$$

where α and β are hyperparameters that control the relative importance of each component.

The Classification Loss is a standard cross-entropy loss, applied to the logits produced by the classifier and the true class labels. It ensures that the model learns to predict the class of the input data correctly.

The Orthogonality Loss, given in the Eq. 2, operates on the latent representations produced by the m network blocks $(\text{latent}_1, \ldots, \text{latent}_m)$. To promote diversity and decorrelation among the extracted features, the orthogonality loss enforces pairwise orthogonality by minimizing the squared cosine similarity between the normalized latent vectors $(\mathrm{u}_1, \ldots, \mathrm{u}_m)$. The orthogonality loss is defined in Eq. 2:

$$\text{Orthogonality Loss} = \frac{1}{N} \sum_{i=1}^{N} \sum_{1 \le j < k \le m} (\mathrm{u}_{j,i} \cdot \mathrm{u}_{k,i})^2 \tag{2}$$

where N is the batch size and $\mathrm{u}_{j,i}$ represents the normalized version of $\text{latent}_{j,i}$, deeply the latent vector for the base block j for the input i, computed as in Eq. 3

$$\mathrm{u}_{j,i} = \frac{\text{latent}_{j,i}}{\|\text{latent}_{j,i}\| + \varepsilon}, \tag{3}$$

with ε being a small constant to prevent division by zero.

The classification loss focuses on the primary task of accurate classification, while the orthogonality loss promotes robustness by reducing redundancy among the latent representations. This added robustness helps the model maintain performance even when inputs are perturbed or missing, making it more stable in real-world scenarios.

Combining these two components in the total loss function (equation 1), the proposed approach achieves improved classification performance while ensuring diversity and robustness in feature extraction.

3. Experimental Results

3.1. Experimental set-up

In this section, we describe the experimental setup used to evaluate the proposed neural network models and their performance under different configurations.

All experiments were conducted on a system equipped with an NVIDIA Tesla T4 GPU. The dataset described in Section 3.2 was split into training, validation, and test sets using a stratified sampling strategy while maintaining a 70-10-20% split. To ensure numerical stability, the following preprocessing steps were applied:

- Input features were normalized to the range $[0, 1]$.
- Missing values (NaNs) were replaced with zero.

The models were trained using the AdamW optimizer due to its efficiency and adaptive learning rate capabilities. The training process was configured as follows:

- Batch size: 4, ensuring stable gradient updates and efficient resource utilization.
- Learning rate: Initially set to 0.0001 to allow controlled parameter updates.
- Maximum epochs: 1000, with early stopping applied to prevent overfitting and unnecessary computations.
- Early stopping: Training was halted if the validation loss did not improve for 10 consecutive epochs.
- Learning rate scheduling: If the validation loss did not improve for 5 consecutive epochs, the learning rate was reduced by a factor of 0.1.

Model hyperparameters were selected using a cross-validation strategy to ensure generalization across different data subsets. The performance of the models was evaluated using classification accuracy, defined as

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total number of samples}}. \tag{4}$$

3.1.1. Ablation Study

To assess the impact of the orthogonality constraint, we conducted an ablation study by training the same models under different weighting schemes for the orthogonality loss. Specifically, we considered two scenarios: one where the orthogonality loss was given equal importance to the classification loss and another where it was entirely omitted. This analysis enabled us to evaluate the effect of enforcing feature diversity through the orthogonality constraint on both classification performance and the model's robustness to input perturbations.

3.2. Dataset and task description

To evaluate the effectiveness of the approach in handling missing modalities, we considered a recently published dataset [23] providing a comprehensive multimodal benchmark for studying behavioral biometrics and sensor-based authentication. The rich multimodal nature of this dataset makes it a valuable test bed for studying missing modality scenarios and the effectiveness of orthogonally constrained loss functions in comparison to traditional approaches. The dataset is publicly available through the European Commission's Joint Research Centre repository[2].

The dataset consists of raw sensor data collected in an office-like environment where 54 adult participants performed six distinct tasks under controlled conditions. Each participant completed two sessions per task, with each task presented in three variations designed to induce different levels of cognitive load. The experimental setup included multiple sensor modalities, such as inertial measurement units (IMUs), force sensors, radar sensors, and electroencephalogram (EEG) recordings. The task aimed to identify each of the 54 individuals based on these biometric sensor data, given the diverse set of physiological and behavioural inputs.

[2] https://data.jrc.ec.europa.eu/dataset/7be86ffd-1bac-4d3e-82fc-02b3ea40ab49

3.3. Experiments on classification tasks

A pre-processing phase identified a total of 46 features extracted from the raw data. Then, all three MHNs are trained on a training set, with each network structured using m equivalent heads (i.e. *m-FFNet*, *m-ConvNet*, and *m-LSTMNet*). We tested different configurations varying $m \in [2, 3, ..., 10]$. This design allows to assess the robustness of different number of heads and different feature extraction paradigms in handling missing features.

After training, we evaluated the models on a separate test set, where all three architectures achieved comparable accuracy scores, consistently falling within a [94%–98%] range. These results indicate that the MHN-based approach can successfully differentiate between the 54 individuals based on the available multimodal biometric signals, demonstrating high classification performance across different feature extraction strategies.

To assess the robustness of the proposed approach through a structured ablation study, we first identified the most influential features contributing to classification accuracy. To this end, we conducted a feature importance analysis using the expected gradients method, as implemented in the SHAP (SHapley Additive exPlanations) python library [24], which assigns each feature an importance value for a particular prediction according to Shapley values [25]. Specifically, we employed SHAP's GradientExplainer to interpret the contribution of each feature to the model's predictions. GradientExplainer is an implementation of expected gradients to approximate SHAP values for deep learning models.

The results of the explainability analysis, as illustrated in Figure 1, reveal that the most significant features $[0, .., 7]$ originate from the electroencephalogram (EEG) signals, one of the key modalities present in the dataset. Thus one modality exhibited the highest contribution to the model's accuracy. This finding also highlights the dominant role of neural activity data in distinguishing among individuals.

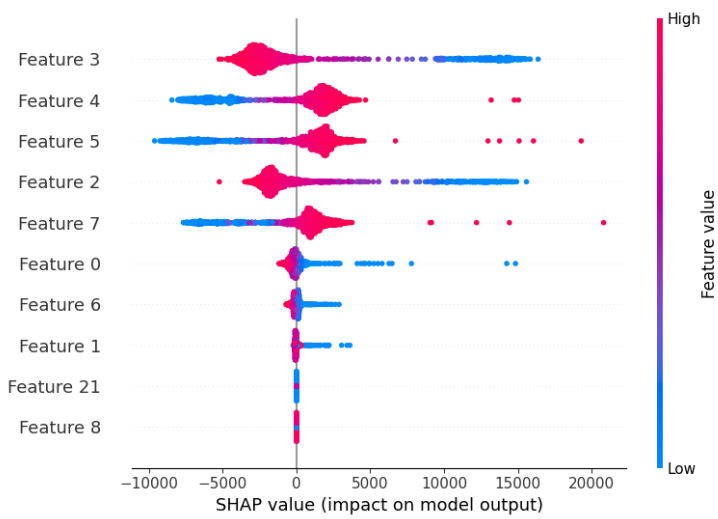

Figure 1. The best 10 informative features according to SHAP's GradientExplainer

These EEG related features are then the one used for the ablation study, whose results are presented in Figure 2, illustrating the impact of orthogonality loss on classification performance across three architectures: *FFNet*, *ConvNet*, and *LSTMNet*. The y-axis of each plot represents the accuracy difference, computed as in Eq. 5

$$\Delta \text{ Accuracy} = \text{Accuracy}_{\text{orthogonal loss}} - \text{Accuracy}_{\text{regular loss}} \tag{5}$$

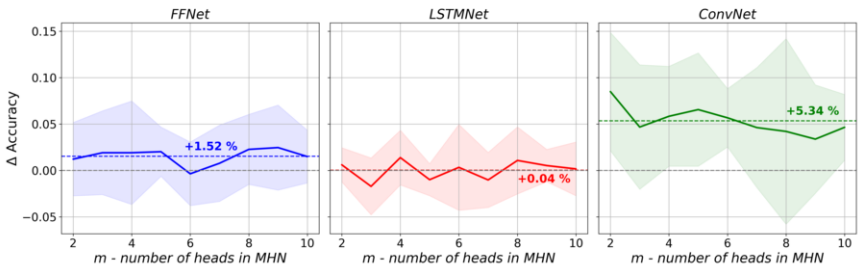

Figure 2. For each number of heads in the MHN (x-axis), the plot shows the average and standard deviation of Δ Accuracy (eq. 5) computed from individual EEG feature ablation experiments. Solid lines represent the average Δ Accuracy per model type, shaded areas indicate one standard deviation, and dotted lines denote the overall average Δ Accuracy across all head configurations for each model.

This value (Eq. 5) quantifies the difference in accuracy between models trained with orthogonality loss and those trained with a regular loss function. Specifically, models were first trained on the full dataset, with all features present, and then tested under conditions where one feature at a time was removed from the most informative modality.

A positive value indicates that the accuracy degradation due to a missing feature is less severe in the MHN trained with orthogonality loss compared to the MHN trained with regular loss. Conversely, a negative value means that the model trained with orthogonality loss experiences a greater accuracy drop than its regular loss counterpart.

The average accuracy difference (+1.52%) in case of *FFNet* suggests that, in this application, the orthogonality loss mitigates accuracy degradation when critical features are missing during testing. However, the results have high variance and show positive and negative variations, indicating that the impact of the orthogonality constraint is inconsistent between different missing feature scenarios.

Unlike the other architectures, results for *LSTMNet* show only marginal improvement (+0.04%), with effects that are largely negligible. While the orthogonality constraint encourages temporal feature diversity, it may also interfere with the recurrent structure's ability to preserve sequential dependencies. These mixed outcomes suggest that orthogonality must be carefully tuned when applied to temporal architectures.

At the other end of the spectrum, *ConvNet* architecture consistently benefit the most from the proposed approach (+5.34% on average). The combination of spatial feature extraction and head-wise orthogonality promotes diverse and complementary representations, enabling the model to better tolerate missing inputs. This confirms that orthogonality is particularly effective when applied to convolutional architectures, likely due to their ability to decompose and generalize local patterns.

It is clear that *ConvNet* exhibits the most pronounced improvement, with an average Δ Accuracy exceeding 5% when critical features are missing. This gain is particularly noteworthy given that all models were trained on the full feature set and evaluated solely under feature ablation scenarios. However, *ConvNet* also shows the widest variance in robustness performance across different heads configurations and different missing features. This variability highlights its potential to achieve exceptionally high improvements (in some cases up to 15%) but also, though rarely, slight degradations.

This behavior underscores a key consideration in applying MHNs with orthogonality regularization: robustness gains must be balanced with stability. The wide variance of *ConvNet* suggests that while the architecture is highly responsive to the benefits of orthogonal heads, it is also more sensitive to their configuration. Consequently, fine-tuning the number of heads is important to ensure consistent performance under missing modality conditions.

In our experiments, the configuration with $m = 6$ heads emerges as the most favorable. It provides a high average robustness gain while simultaneously reducing variability compared to other configurations.

These results suggest that architectural redundancy, achieved through multiple independently parameterized heads, can significantly improve resilience to missing features, provided the number of heads is carefully tuned to avoid overparameterization and instability. Overall, our findings validate the proposed approach as a viable and effective strategy for enhancing robustness in multimodal learning systems.

4. Conclusions and perspective outlooks

This study investigates the integration of Multi-Head Network (MHN) architectures with orthogonality-based regularization as a strategy to enhance robustness against missing modality features in multimodal learning. Our results show that the degree of improvement is architecture-dependent, with convolutional models benefiting the most from the proposed approach.

To further validate our approach, future research will extend evaluation to additional datasets and application domains, aiming to assess its adaptability in diverse multimodal contexts. A refined training and testing pipeline is also under development: after training MHNs with heterogeneous heads, explainability techniques will identify the most informative modalities. This will guide the selection and tuning of the number of heads to maximize robustness while minimizing performance variability when critical features are missing.

The code for our experiments will be made publicly available to foster further research and development in multimodal AI.

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